



Upstream Reservoir Cascades Rewire Changjiang Carbon

A staircase of reservoirs on the upper Changjiang has rearranged where river-borne carbon comes to rest. Ke and colleagues follow particulate organic carbon (POC) from mountain headwaters to the tidal limit using grain size, major elements, and dual carbon isotopes (^{13}C , ^{14}C) to see what dams change—and what they preserve.

Their verdict: cascade mega-reservoirs now sequester about 6.6 MtC yr^{-1} of POC, including 3.8 MtC yr^{-1} from the modern biosphere. The Three Gorges Dam's POC burial fell from 1.6 to 0.4 MtC yr^{-1} as new upstream reservoirs began storing roughly 0.7 MtC yr^{-1} , relocating burial from the estuary to the Jinshajiang cascade. Faster burial in high-relief reservoirs shortens oxygenated transit, curbing mineralization and boosting preservation.

Meanwhile, downstream exports grow relatively richer in biospheric POC but smaller in total load—reshaping the East China Sea's carbon budget and, at times, tipping mid-Changjiang reaches from sink to source.

The study reframes dams as more than sediment traps: they rewire carbon pathways and ages, filtering petrogenic fragments and accelerating storage of terrestrial carbon on land. For Critical Zone science and carbon accounting, the message is clear—engineering choices set the tempo of erosion, transit, and burial across one of Earth's great river systems.

Ke, Y., Calmels, D., Bouchez, J., Noret, A., Massault, M., Chetelat, B., et al. (2025). Regulation of particulate organic carbon by cascade mega-reservoirs in the Changjiang basin: Enhanced sequestration and altered downstream composition. *Global Biogeochemical Cycles*, 39, e2024GB008479.

“Quantifying the long-term impact of dam-induced dynamics on POC preservation and metabolism will be crucial for understanding how human behaviors regulate carbon cycling and climate change.”

